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Unit 5 : Session 10 : SLO 2 | ARM Addressing Modes

I. Multiple Choice

1. C) Pre-indexed the offset is applied to the base register before the memory access.
2. B) LDR R0, =value the pseudo-instruction loads a constant from the literal pool.
3. B) Access memory post-indexed accesses memory at the old base first, then updates the register.

II. Fill in the Blanks

1. register register-offset addressing uses a register operand as the offset.
2. literal literal addressing loads constants from the literal pool.
3. after post-indexed addressing updates the base register after accessing memory.

III. True or False

1. TRUE.
2. FALSE LDR/STR do support scaled index addressing (e.g. [R1, R2, LSL #2]).
3. TRUE the literal pool is accessed PC-relative.

IV. Match the Following

- A) Immediate Offset 3) Uses immediate displacement.
B) Pre-indexed 2) Offset added before access.
C) Post-indexed 4) Base updated after access.
D) Literal Addressing 1) Uses PC-relative constant.

V. Numerical Answers

1. $0x200C = 0x2000 + 12 = 0x200C$.
2. $0x1010 = R2 \text{ LSL } \#2 = 4 \times 4 = 16 = 0x10$; $0x1000 + 0x10 = 0x1010$.
3. $0x3004$ post-indexed: memory is accessed at $0x3000$, then $R1 = 0x3000 + 4 = 0x3004$.